$\gamma(h)$ for the MA(q) Process

Suppose that $h \geq 0$

Consider

$$\gamma(h) = \operatorname{Cov}(X_t, X_{t+h})$$

$$= \mathbb{E}(X_t X_{t+h}) - \mathbb{E}(X_t)\mathbb{E}(X_{t+h})$$

$$= \mathbb{E}(X_t X_{t+h}) \quad \text{because } \mathbb{E}(X_t) = 0 \text{ for all } t.$$

$$= \mathbb{E}\left[\underbrace{\left(\sum_{j=0}^{q} \theta_j Z_{t-j}\right)}_{X_t}\underbrace{\left(\sum_{k=0}^{q} \theta_k Z_{t+h-k}\right)}_{X_{t+h}}\right]$$

$$= \sum_{j=0}^{q}\sum_{k=0}^{q} \theta_j \theta_k \mathbb{E}(Z_{t-j} Z_{t+h-k}) \quad (+)$$

Consider $\mathbb{E}(Z_{t-j} Z_{t+h-k})$ - this is equal to 0 <u>EXCEPT</u> when $-j = h - k$

$$\Rightarrow k = j + h$$

$\cancel{\therefore \gamma(h) =}$

Consider different cases.

① <u>$h > q$</u>

If $h > q$ then if $k = j + h \Rightarrow k > q$

$k \in \{0, \dots, q\}$

If $\gamma(h)$ is non-zero then h is not greater than q.

For a non-zero sum in $(+)$ we require

$j+h \in \{0, \dots, q\}$

<u>and</u>

$j \in \{0, \dots, q\}$

$\therefore \; j \in \{0, \dots, q-h\}$

then $(+)$ becomes

$$\gamma(h) = \sum_{j=0}^{q-h} \theta_j \theta_{j+h} \mathbb{E}(Z_{t-j}^2)$$

$$= \sigma_Z^2 \sum_{j=0}^{q-h} \theta_j \theta_{j+h}.$$

for $0 \leq h \leq q$

$\operatorname{Var}(Z_{t-j})$
$= \mathbb{E}(Z_{t-j}^2) - \mathbb{E}(Z_{t-j})^2$
$= \mathbb{E}(Z_{t-j}^2)$
$\because \mathbb{E}(Z_{t-j}) = 0$

We can use a similar approach to determine $\gamma(-h)$ and deduce that $\gamma(h) = \gamma(-h)$

$$\therefore \; \gamma(h) = \begin{cases} 0 & h > q; \\ \sigma_Z^2 \sum_{j=0}^{q-h} \theta_j \theta_{j+h} & 0 \leq h \leq q; \\ \gamma(-h) & h < 0. \end{cases}$$